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VPC Network Access Analyzer

VPC Network Access Analyzer

This section of the workshop demonstrates how Amazon VPC Network Access Analyzer helps you identify and assess potential security risks by analyzing network configurations that could allow unintended network access across your VPC infrastructure.

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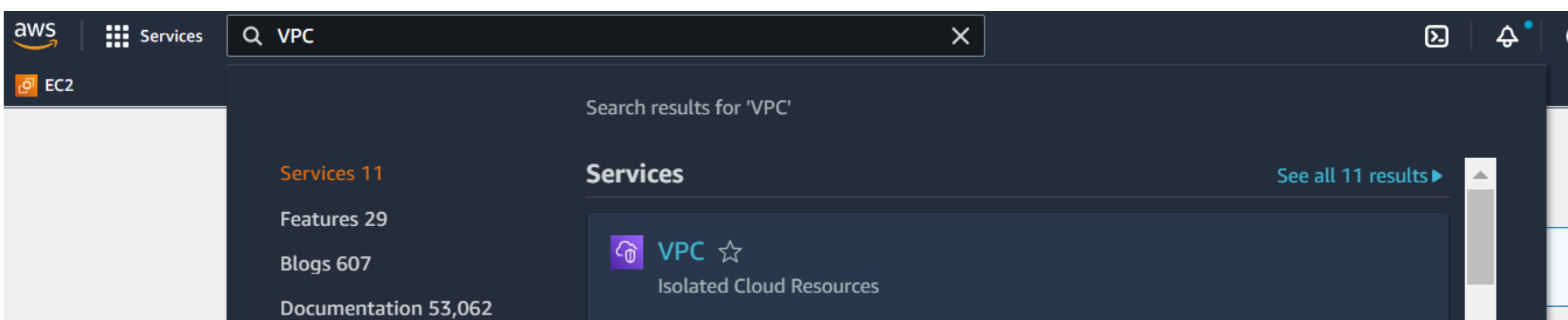
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Lab 1 - Set up Amazon VPC Network Access Analyzer and check Internet accessibility

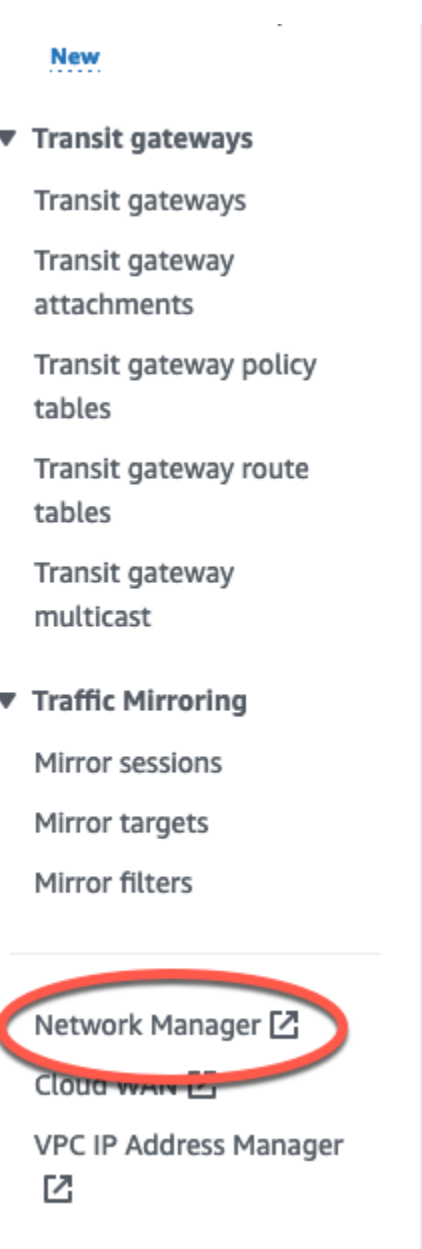
Objective: In this lab, you will use Amazon VPC Network Access Analyzer to identify resources that are accessible from the internet. This analysis will help verify that only resources requiring legitimate internet access are exposed through internet gateways, enhancing your security posture.

Enable Amazon VPC Network Access Analyzer

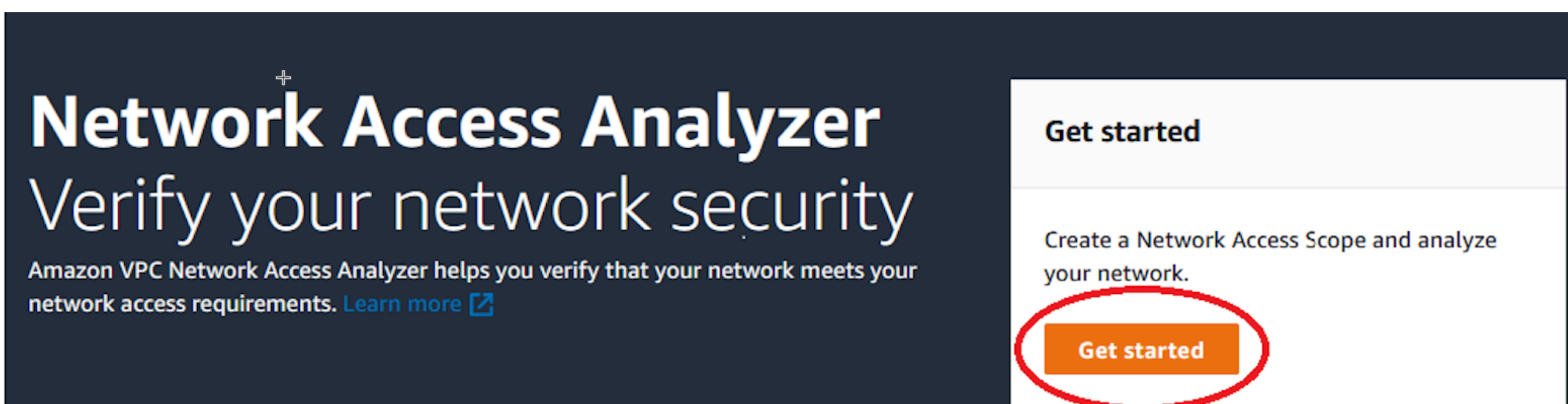
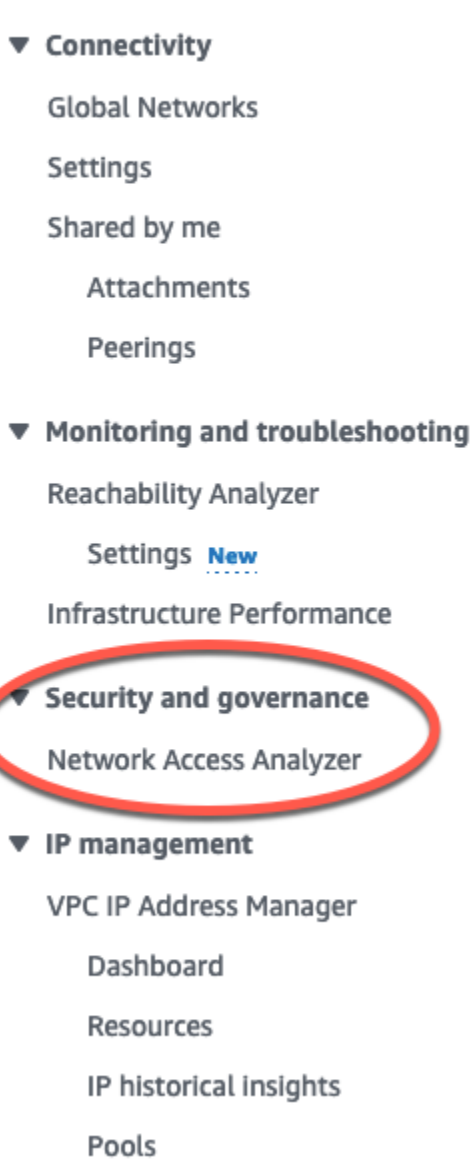
1. From the **AWS Console** click in the search bar in the top left corner. Type **VPC** in the services search bar. Select **VPC** from the list.



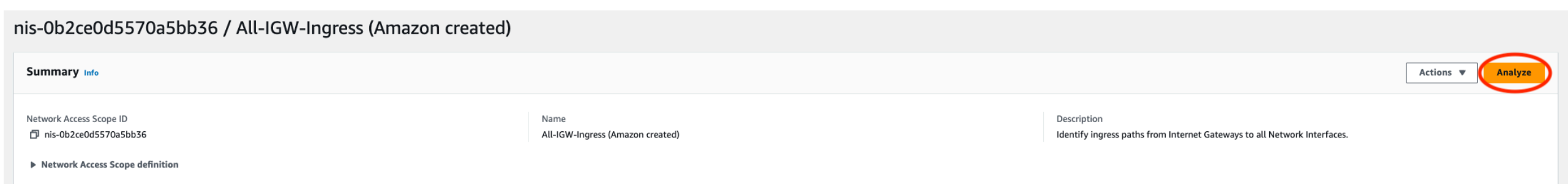
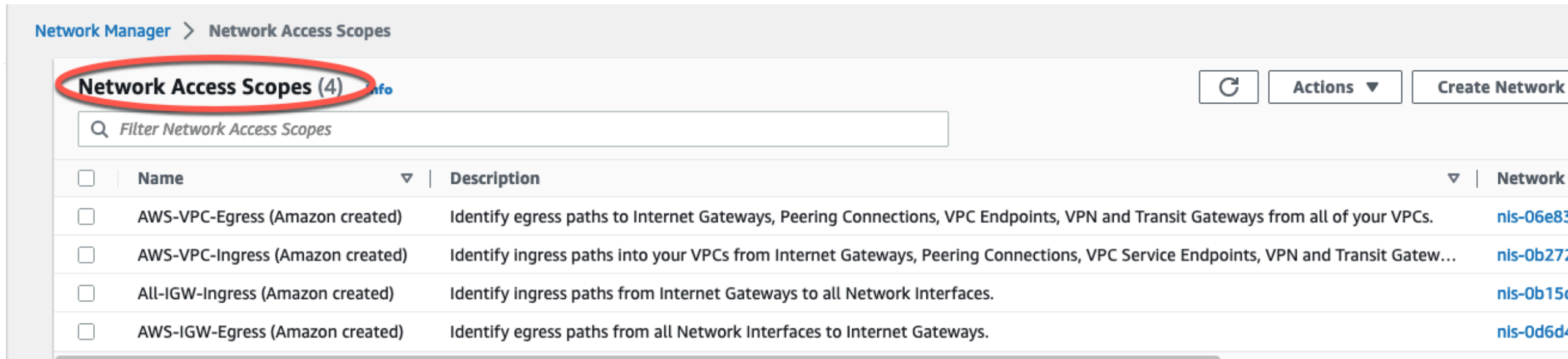
2. Scroll on the left pane and Click **Network Manager** on the left side of the screen. A new browser tab will open up. Click on **Network Access Analyzer** which can be found under the **Security and governance** on the left pane and then click **Get started**. Please refer the below screenshots.



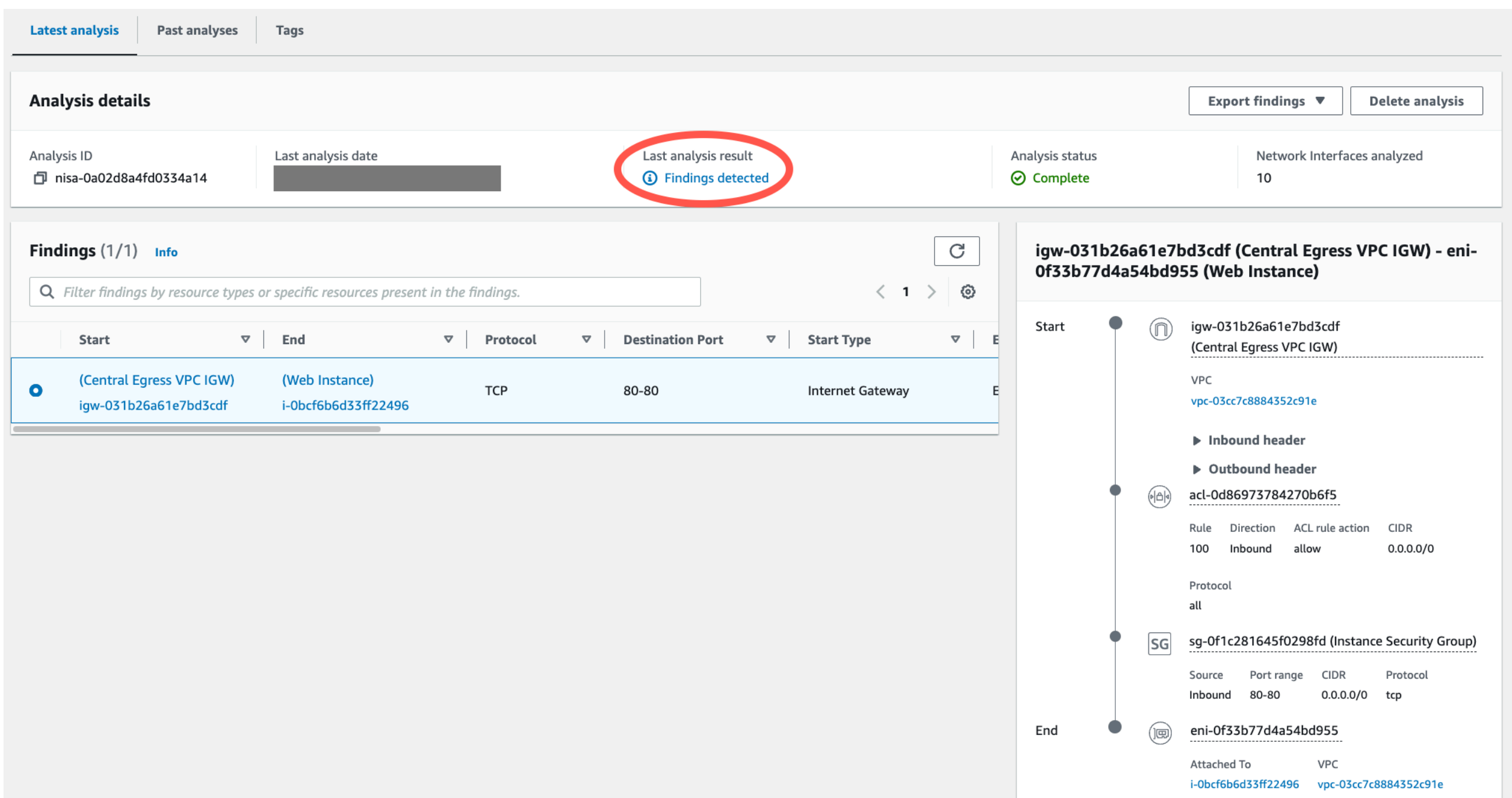
AWS Network Manager



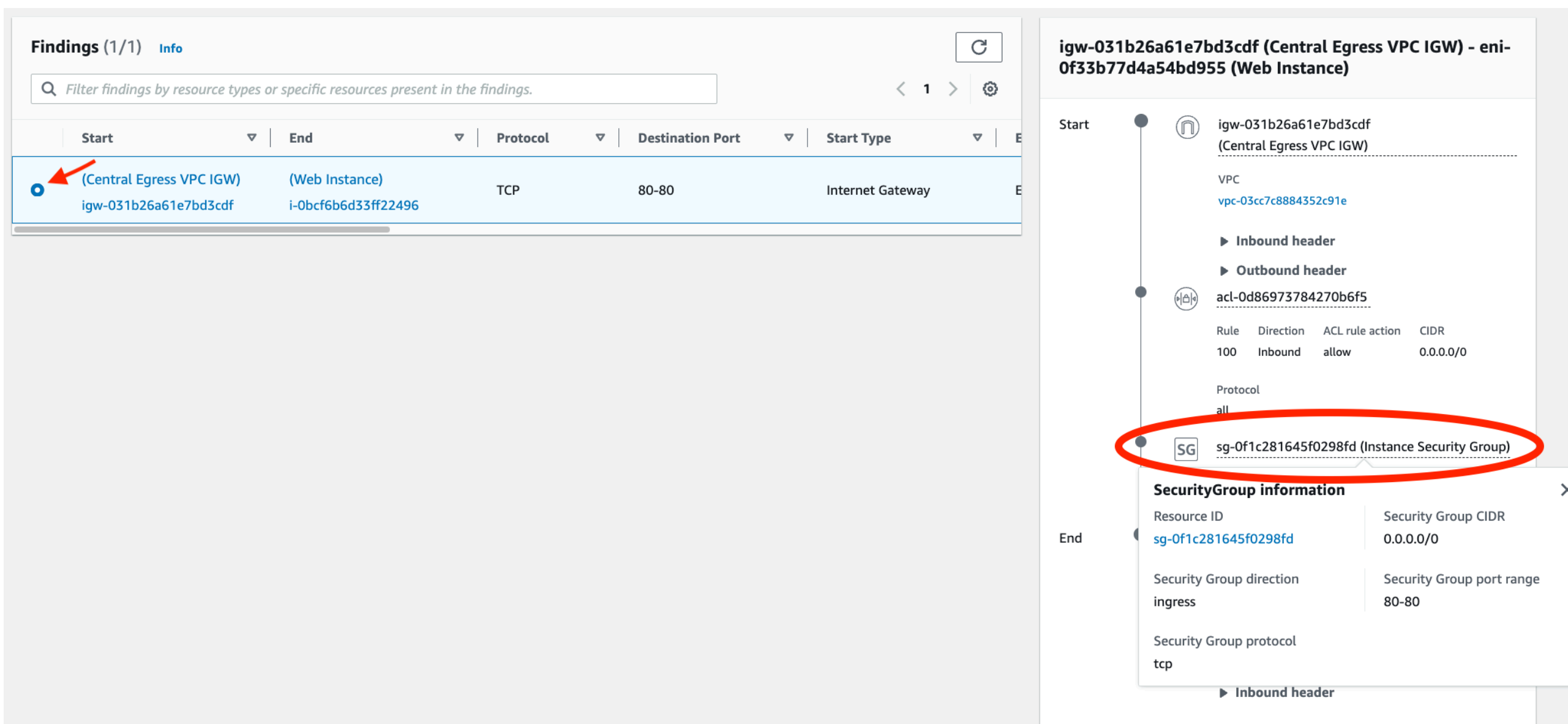
3. You will see 4 scopes which were created by Amazon.



5. Scroll down to explore the findings once complete. Each finding shows a network path from internet through internet gateway to the resources in AWS(example: EC2 instances). This is where you would review each finding and flag/take action for findings/paths that were not intended to have access for the internet.



6. Select the first finding by clicking the radio button and in the right pane you can click on a resource to see additional information on a given resource.

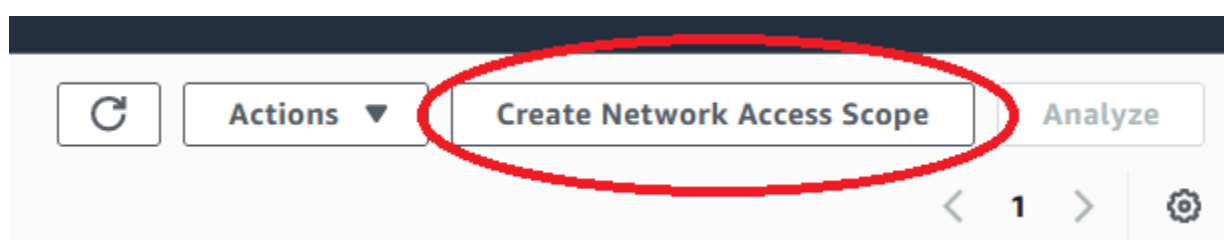


Summary: In this lab, you learned how to enable Amazon VPC Network Access Analyzer and used a default Amazon network access scope to analyze ingress traffic paths from the internet to resources in your VPC.

Lab 2 - Network segmentation via AWS Transit Gateway

Objective: In this lab, we will examine network segmentation across multiple Virtual Private Clouds (VPCs) connected through AWS Transit Gateway. Our primary objective is to create two custom network access scopes: the first to verify that Production and Development VPCs are properly isolated and cannot communicate with each other, and the second to confirm that the Development VPC can successfully communicate with the Inspection VPC, as required by the network design.

- Click on **Network Access Analyzer** on the left side to get back to the main page. Click **Create Network Access Scope** in the upper right.



- Select **Validate network segmentation** radio button and click **Next**.

Network Access Scope templates

Select Network Access Scope template

Select template

Build your Network Access Scope starting from a template based on common network access scenarios.

☐ Identify access from internet gateways

Example

- Locate databases accessible from internet
- Find non-HTTPS access to web servers

☐ Identify access to internet gateways

Example

- Locate instances with un-authorized internet access

☐ Validate access from trusted networks

Example

- Containers can only be accessed via load balancers
- Only Bastions can SSH to production
- Only App Servers can access Database Servers

☒ Validate network segmentation

Example

- Development should be isolated from Production
- PCI should be isolated from Non-PCI

☐ Assess impact of VPC Block Public Access

Example

- Locate servers accessible from the Internet
- Locate resources with Internet access

☐ Empty template

Build your own Network Access Scope

Cancel

Next

- Provide an optional Name and Description, such as "Lab2," to easily identify it.

- Under **Match conditions - Match findings - condition 1**:

- For the Source, select **Resource IDs** under Resource selection, select **Transit Gateway Attachments** under Resource types and select **Dev VPC TGW Attachment** under Resource IDs.
- For the Destination, select **Resource IDs** under Resource Selection, select **Transit Gateway Attachments** under Resource types and select **Prod VPC TGW Attachment** under Resourceas IDs as shown below.

Source

Resources

Resource selection

Resource IDs

Resource types

Select resource types

Transit Gateway Attachments

Resource IDs

Select resource IDs

tgw-attach-0c02681ab67e06dea (Dev VPC TGW Attachment)

Traffic type- optional

Destination

Resources

Resource selection

Resource IDs

Resource types

Select resource types

Transit Gateway Attachments

Resource IDs

Select resource IDs

tgw-attach-09ba0ce6b2z848c582 (Prod VPC TGW Attachment)

Traffic type- optional

- Click **Next** in the lower right and on the Review and create screen, click **Create Network Access Scope** in the lower right.

- Select the **Network Access Scope** (i.e. Lab2) and click **Analyze** in the upper right.

- Once complete you can see that there are no findings which confirms our VPC's - Production and Development are segmented and traffic cannot route between them.

Analysis details

Analysis ID

nis-a0af2fe6215act54

Last analysis date

Last analysis result

No findings detected

Analysis status

Complete

Network interfaces analyzed

10

Delete analysis

Next, let's create a new network access scope to verify if traffic from the Development VPC can reach the Network Firewall in the Inspection VPC. To simplify this process, we can clone our previously created network access scope and modify it for this analysis.

- Click on **Actions** in the right hand corner and select **Duplicate and modify** as shown below.

VPC > Network Access Scopes > nis-0315e7674479d5b89

nis-0315e7674479d5b89 / Lab2

Summary

Info

Network Access Scope ID

nis-0315e7674479d5b89

Name

Lab2

Description

-

Network Access Scope definition

Actions

Analyze

Duplicate and modify

Manage tags

Delete Network Access Scope

- Provide an optional Name and Description such as **Lab2a** to easily identify it.

- For the Destination, deselect **Prod VPC TGW Attachment** and select **Inspection VPC TGW Attachment** under Resource IDs as shown below.

Source

Resources

Resource selection

Resource IDs

Resource types

Select resource types

Transit Gateway Attachments

Resource IDs

Select resource IDs

tgw-attach-0c02681ab67e06dea (Dev VPC TGW Attachment)

Traffic type- optional

Destination

Resources

Resource selection

Resource IDs

Resource types

Select resource types

Transit Gateway Attachments

Resource IDs

Select resource IDs

tgw-attach-00567af5b0547a467 (Inspection VPC TGW Attachment)

Traffic type- optional

- Click on **Duplicate and analyze Network Access Scope**.

- The analysis reveals expected findings, confirming that resources in the Development VPC can route traffic through the Network Firewall in the Inspection VPC.

Analysis details

Analysis ID

nis-a03237c4815fdd8dfe

Last analysis date

Last analysis result

Findings detected

Analysis status

Complete

Network interfaces analyzed

10

Export findings

Delete analysis

Findings (1/14)

Filter findings by resource types or specific resources present in the findings.

Start	End	Protocol	Destination Port	Start Type	End Type
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	TCP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	UDP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	TCP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	TCP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	UDP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	UDP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	TCP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	UDP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	TCP	0-65535	Transit Gateway	Transit Gateway
tgw-0782a4bf52b2f8118	tgw-0782a4bf52b2f8118	UDP	0-65535	Transit Gateway	Transit Gateway

tgw-attach-09923c4dc1f00a2c9 (Dev VPC TGW Attachment)

Transit Gateway ID

tgw-0782a4bf52b2f8118

Outbound header

tgw-rtb-056b72a06932ba9c7 (Dev Inspection Route Tab...)

Destination CIDR

State

0.0.0.0/0

active

Attachment ID

tgw-attach-0293a4b799d115ea5

Resource ID

vpc-0c5a5dbcbba15fa4b8

tgw-attach-0293a4b799d115ea5 (Inspection VPC TGW Atteac...)

Transit Gateway ID

tgw-0782a4bf52b2f8118

Inbound header

Summary: In this lab, you successfully created a network access scope to analyze VPC segmentation and confirmed the proper isolation between Production and Development VPCs. Through the analysis, you verified that the network architecture adheres to the intended security design by ensuring these environments remain separated.

In another network access scope analysis, you verified that traffic from the Development VPC can properly route to the Inspection VPC, which aligns with the intended network design. It's important to note that if the analysis had revealed any unintended communication paths between VPCs that should be isolated, these would be flagged as findings.

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Accept

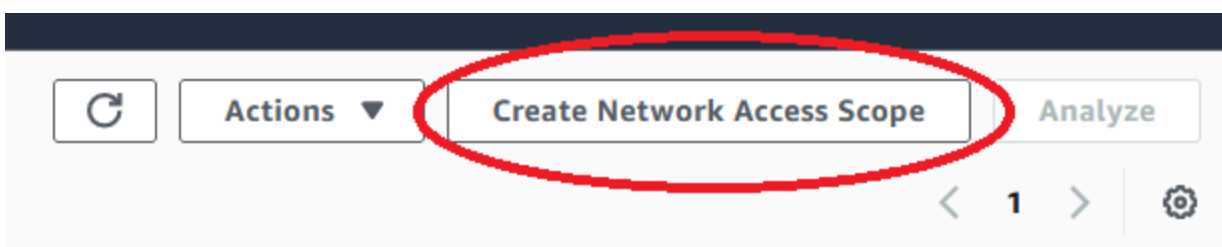
Decline

Customize

Lab 3 - Security controls (e.g., firewall/NAT-GW) in path

Objective: AWS Network Access Analyzer helps verify that your network traffic follows designated paths through security controls like network firewalls and NAT gateways. In this lab, you will create custom network access scopes to verify that instances in both Production and Development VPCs properly route internet bound traffic through the Network Firewall deployed in the Inspection VPC.

- Click on **Network Access Analyzer** on the left side to get back to the main page. Click **Create Network Access Scope** in the upper right.



- Select **Empty template** radio button and click **Next**.

Network Access Scope templates

Select Network Access Scope template

Select template
Build your Network Access Scope starting from a template based on common network access scenarios.

☐ **Identify access from internet gateways**
Example

- Locate databases accessible from internet
- Find non-HTTPS access to web servers

☐ **Identify access to internet gateways**
Example

- Locate instances with un-authorized internet
 - access

☐ **Validate access from trusted networks**
Example

- Containers can only be accessed via load balancers
- Only Bastions can SSH to production
- Only App Servers can access Database Servers

☐ **Identify non-permissible traffic type**
Example

- Only Web servers can receive HTTP/HTTPS traffic
- Production servers cannot send SSH/RDP traffic
- Development cannot SSH to Production

☐ **Validate network segmentation**
Example

- Development should be isolated from Production
- PCI should be isolated from Non-PCI

☐ **Assess impact of VPC Block Public Access**
Example

- Locate servers accessible from the Internet
- Locate resources with Internet access

☒ **Empty template**
Build your own Network Access Scope

Cancel

Next

- Provide an optional Name and Description such as **Lab3** to easily identify it.

- Under **Match conditions** , click **Add match condition**.

- For the Source, Select **Resource IDs** under Resource selection and select **VPCs** under Resource types. Select the **Prod VPC** and **Dev VPC** under Resource IDs as shown below.

Source

▼ Resources

Resource selection

Resource IDs

Resource types

Select resource types

VPCs

Resource IDs

Select resource IDs

vpc-00a6a5680dc3a9fbf (Prod VPC) vpc-06a63dd22787edfc8 (Dev VPC)

► Traffic type- optional

- Under Destination, select **Resource types** under Resource selection and select **Internet Gateways**. Select the **Central Egress VPC IGW** under Resource IDs as shown below.

Destination

▼ Resources

Resource selection

Resource IDs

Resource types

Select resource types

Internet Gateways

Resource IDs

Select resource IDs

igw-Offe99a0e7ae5b53c (Central Egress VPC IGW)

► Traffic type- optional

Add match condition

- Click **Next** in the lower right and on the Review and create screen, click **Create Network Access Scope** in the lower right.
- Select the **Network Access Scope** (i.e. **Lab3**) and click **Analyze** in the upper right.
- Once complete you can see that there are findings. We are interested in looking at the 'TCP' traffic so we can filter the findings as shown in the diagram.

Analysis details

Analysis IDnisa-0b24206954cb3846

Last analysis date

Last analysis resultFindings detected

Analysis statusComplete

Network Interfaces analyzed9

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Findings (1/2) Info

Filter findings by resource types or specific resources present in the findings.

Protocol: TCP Clear filters

Start	End	Protocol	Destination Port	Start Type
(Dev Instance) i-0303f01ed38dda7f9	(Central Egress VPC IGW) igw-077d240b498d3ac07	TCP	0-65535	EC2 Instance
(Prod Instance) i-00f9c045197443388	(Central Egress VPC IGW) igw-077d240b498d3ac07	TCP	0-65535	EC2 Instance

eni-00f5e703cac42fd34 (Dev Instance) - igw-077d240b498d3ac07 (Central Egress VPC IGW)

Start

eni-00f5e703cac42fd34
Attached To
i-0303f01ed38dda7f9 (Dev Instance)
VPC
vpc-0c329de39654596ad (Dev VPC)
Subnet
subnet-0101476e0e5af557d (Dev Workload Subnet)
▼ Outbound header
Destination address
0.0.0.0/5
Destination port range
0-65535
Protocol
TCP
Source address
10.2.1.4/32
Source port range
0-65535
sg-03e32a0667a4f203c (Dev Instance Security Gr...)
Destination
0.0.0.0/0
Outbound
all
Protocol
all
acl-03994d4042910b4fa
Rule
200
Direction
Outbound
ACL rule action
allow
CIDR
0.0.0.0/0

You will see that the Prod and Dev VPCs have a route through the network firewall (Inspection VPC). Note that you can filter the findings using resource type, protocol etc. Select one of the instances. You may notice the destination address to quad zero i.e. route to the internet.

Scroll down in the right pane you will see the complete path ending at Central Egress VPC IGW. Notice the network firewall in the path as shown below.

To identify any network paths that bypass the Network Firewall and potentially violate compliance requirements, let's create a new network access scope and analyze it. To simplify this process, we can clone our previously created network access scope and modify it for this analysis.

- Click on **Actions** in the right hand corner and select **Duplicate and modify**.

- Provide an optional Name and Description such as **Lab3a** to easily identify it.
- Scroll down to **Exclusion Conditions**, click on **Add exclusion condition** and add **AWS Network Firewalls** in the Through section under Resource types as shown below.

Exclusion conditions - optional Info

▼ Exclusion findings - condition 1

Remove exclusion condition

Source

▼ Resources

Resource selection

Select resource

► Traffic type- optional

Destination

▼ Resources

Resource selection

Select resource

► Traffic type- optional

Through

▼ Resources

Resource selection

Resource types

Resource types

Select resource types

AWS::NetworkFirewall::Firewall

Add exclusion condition

- Click on **Duplicate and analyze Network Access Scope**.

- The analysis shows no findings, which confirms the expected configuration: all traffic from both Production and Development VPCs is correctly routing through the Network Firewall, adhering to compliance requirements.

Summary: In this lab, you used Network Access Analyzer to verify that both Production and Development VPCs correctly route their internet-bound traffic through the Network Firewall. By creating an additional network access scope with specific exclusions, you learned how to identify any potential bypass routes that circumvent the Network Firewall.

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Accept

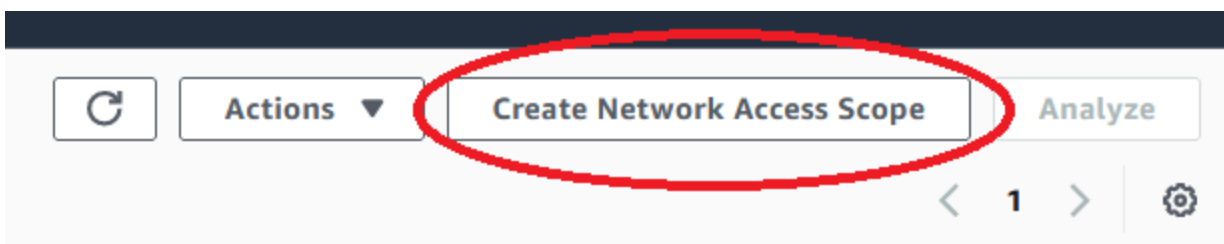
Decline

Customize

Lab 4 - Trusted network access

Objective: AWS Network Access Analyzer helps ensure your resources communicate only with authorized endpoints using specified protocols and ports. In this lab, you will create a custom network access scope to verify compliance with the following requirement: Production instance outbound internet traffic must be restricted to port 443, accessing only the trusted IP range 55.3.0.0/16 for system updates.

- Click on **Network Access Analyzer** on the left side to get back to the main page. Click **Create Network Access Scope** in the upper right.



- Select **Identify access to internet gateways** radio button and click **Next**.

Select Network Access Scope template

Select template

Build your Network Access Scope starting from a template based on common network access scenarios.

☐ **Identify access from internet gateways**
Example

- Locate databases accessible from internet
- Find non-HTTPS access to web servers

☒ **Identify access to internet gateways**
Example

- Locate instances with un-authorized internet access

☐ **Validate access from trusted networks**
Example

- Containers can only be accessed via load balancers
- Only Bastions can SSH to production
- Only App Servers can access Database Servers

☐ **Identify non-permissible traffic type**
Example

- Only Web servers can receive HTTP/HTTPS traffic
- Production servers cannot send SSH/RDP traffic
- Development cannot SSH to Production

☐ **Validate network segmentation**
Example

- Development should be isolated from Production
- PCI should be isolated from Non-PCI

☐ **Assess impact of VPC Block Public Access**
Example

- Locate servers accessible from the Internet
- Locate resources with Internet access

☐ **Empty template**
Build your own Network Access Scope

- Provide an optional Name and Description such as **Lab4** to easily identify it.

- Under **Match conditions - Match findings - condition 1**:

- For the Source, select **Resource IDs** under Resource selection and select **EC2 Instances** under Resource types. Select the instance which has a name of (**Prod Instance**) under Resource IDs.
- The Destination selection is already done.

Match conditions

▼ Match findings - condition 1 Remove match condition

Source

▼ Resources

Resource selection

Resource IDs

Resource types

Select resource types

EC2 Instances

Resource IDs

Select resource IDs

I-00F9c045197443388 (Prod Instance)

► Traffic type- optional

Destination

▼ Resources

Resource selection

Resource types

Select resource types

AWS::EC2::InternetGateway

► Traffic type- optional

Add match condition

- In the Exclusion conditions section, click **Add exclusion condition** and expand the **Traffic type** section under Destination as shown below.

- Enter 55.3.0.0/16 for the Destination address.
- Enter 443 for the Destination port.
- Select TCP for the Traffic type.

Exclusion findings - condition 1

Remove exclusion condition

Source

▼ Resources

Resource selection

Select resource

► Traffic type- optional

Destination

▼ Resources

Resource selection

Select resource

▼ Traffic type- optional

Source address

Enter IP addresses or CIDRs.

10.0.0.0, 10.0.0.0/24

Destination address

Enter IP addresses or CIDRs.

10.0.0.0, 10.0.0.0/24

55.3.0.0/16

Source prefix lists

Select prefix lists

Destination prefix lists

Select prefix lists

Source port

Enter port or port range. Example 443 or 53-80.

Any

Destination port

Enter port or port range. Example 443 or 53-80.

443

Traffic type

TCP

- Click **Next** in the lower right and on the Review and create screen, click **Create Network Access Scope** in the lower right.

- Select the **Network Access Scope (i.e. Lab4)** and click **Analyze** in the upper right.

- The analysis reveals findings that indicate network paths exist between the EC2 instance and the internet gateway beyond our defined restrictions. This is because the current security group configuration allows outbound traffic to all destinations (0.0.0.0/0) over port 443, which is broader than our intended policy of limiting traffic to a specific IP range and port.

Select the first finding by clicking the radio button and in the bottom pane that opens up choose the security group and click on the corresponding resource id starting with 'sg-'.

Analysis ID

nisa-090469a47f8bf3bdc

Last analysis date

Last analysis result

Findings detected

Analysis status

Complete

Network Interfaces analyzed

10

Findings (1/1)

Filter findings by resource types or specific resources present in the findings.

Start	End	Protocol	Destination Port
☒	(Prod Instance) i-0a85a590d03416af0	(Central Egress VPC IGW) igw-03a404a39e8bb5262	TCP 443-443

eni-0851f11ced4f614d2 (Prod Instance) - igw-03a404a39e8bb5262 (Central Egress VPC IGW)

SecurityGroup information

Resource ID

sg-0b9ea095c2e2a79d8

Security Group direction

egress

Security Group protocol

tcp

Security Group CIDR

0.0.0.0/0

Security Group port range

443-443

sg-0b9ea095c2e2a79d8 (Instance Security Group)

Destination

Port range

CIDR

Outbound

443-443

0.0.0.0/0

Protocol

tcp

- In the security group page, choose **Outbound rules** and then **Edit outbound rules**. Now let us tighten the security group to only allow HTTPS traffic to 55.3.0.0/16 over port 443 as shown in the figure below and choose **Save rules**.

VPC

>

Security Groups

>

sg-07a246575ef4148bf - Prod Instance Security Group

>

Edit outbound rules

Edit outbound rules

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules

Security group rule ID

sg-0584b493abdc0157ca

Type

HTTPS

Protocol

TCP

Port range

443

Destination

Custom

Description - optional

Delete

Add rule

- Go back to the **Network access analyzer**, select the network access scope you created for this lab and select **Analyze**. Once complete you can see that there are no findings which confirms that we are in compliance. (i.e. Our ec2 instance can only download updates from 55.3.0.0/16 over TCP port 443).

Summary: In this lab, you used Network Access Analyzer to examine outbound internet traffic paths from an EC2 instance. Through initial analysis and subsequent security

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Accept

Decline

Customize